Julio de Mello Bezerra

Mathematics PhD | Cryptography & Cryptanalysis

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Location: Regensburg, Germany • **Nationality:** Portuguese (EU) / Brazilian

Profile

PhD mathematician specializing in number theory and arithmetic geometry, with a broad theoretical command of classical and post-quantum cryptography. Experienced in analyzing cryptographic constructions and cryptanalytic techniques, independently validating mathematical arguments, identifying algorithmic improvements and potential security weaknesses, and turning research literature into computational experiments using Python and SageMath. Particularly adept at rapidly mastering new technical domains and translating complex theoretical concepts into actionable security analysis for real-world cryptographic systems and protocols.

Skills & Languages

Cryptographic Focus: Elliptic-curve & pairing-based cryptography, zk-SNARKs, cryptanalysis, primality testing and number-field sieve algorithms.

Additional Knowledge: RSA and AES, lattice-based and isogeny-based post-quantum cryptography.

Programming & Tools: Python, SageMath, LaTeX, basic familiarity with Linux & Git.

English	Fluent
German	Working Proficiency (University teaching experience)
Portuguese	Native
Spanish	Basic Proficiency

Cryptographic Research & Projects

Yukon CurveSmash Autoresearch Project (yukon.org/curvesmash) **2026 – Present**
Judge & Cryptanalysis Researcher *Remote*

- **Multi-Curve Cryptanalysis:** Independently evaluated and mathematically verified AI-assisted cryptanalytic approaches against widely used elliptic curves (BLS12-381, BN254, secp256k1 & secp256r1). Compared proposed attacks against existing literature, audited attacks strategies and identified improved parameters.
- **Security of Pairing-Friendly Curves (Article):** Authored an interactive technical analysis of the SexTNFS attack, comparing CurveSmash's advancements with the existing state of the art in the literature. Refined attack-cost estimates and reduced the expected security of pairing-friendly curves, specifically BLS12-381 & BN254.
- **zkSNARK Setup Security (Article):** Co-authored *Beyond the Cheon-Kozaki-Kutsuma-Matsuo attack*, analyzing the algebraic and algorithmic ideas behind new discrete-logarithm speedups that exploit "powers-of-tau" auxiliary inputs, often used in KZG commitment schemes.

Applied & Theoretical Cryptography **2026**
Projects & Study *Independent*

- **From Research Papers to Optimized Code (Article & Code):** Authored an interactive article and Python/SageMath implementation on adversarial primality testing (forging composites that fool naive Miller–Rabin prime checks), translating research papers into optimized executable code and achieving orders-of-magnitude speedups via single-pass CRT alignment, segmented sieving and quadratic reciprocity.
- **CTF Challenges:** Solved over 70 Capture-the-Flag challenges on cryptohack.org (reaching the top 1% of users) by implementing practical attacks in Python and SageMath across most standard symmetric and asymmetric schemes, such as AES in various modes, RSA, Diffie-Hellman, ECC and more.
- **Post-Quantum Cryptography:** Studied quantum cryptanalytic algorithms and isogeny-based protocols as a participant in the QuSoft reading group on *Quantum Algorithms for Isogenies* (Spring 2021), as well as the subsequent higher-dimensional attack on SIDH. Studied lattice-based post-quantum cryptography through ML-KEM (CRYSTALS-Kyber), including its role in hybrid key-exchange protocols in TLS 1.3.

Professional Experience

University of Regensburg **Oct 2020 – Jul 2026**
Doctoral Researcher & Lecturer *Regensburg, Germany*

- **Research:** Conducted independent research in number theory and arithmetic geometry resulting in a *magna cum laude* doctoral thesis, alongside collaborative research in advanced departmental seminars (*Oberseminare*).
- **Teaching:** Instructed undergraduate and graduate courses across 8 semesters in English and German, covering general mathematics as well as advanced seminars in cryptography-relevant topics including Elliptic Curves and Lattices.
- **Conferences:** Co-organized a workshop on Eisenstein series and equivariant cohomology, delivered research talks in Germany, France, and the UK, and participated in multiple international conferences across Europe.

Education

University of Regensburg **Apr 2026**
Doktor der Naturwissenschaften (PhD in Mathematics) – magna cum laude *Regensburg, Germany*

- **Thesis:** *Shintani cocycle decomposition of topological polylogarithms*

University of Regensburg & Leiden University **Aug 2020**
ALGANT Master in Mathematics (Scholarship Recipient) *Germany & Netherlands*

Universidade Federal do Rio de Janeiro (UFRJ) **Jan 2018**
Bachelor in Pure Mathematics *Rio de Janeiro, Brazil*

Selected Technical Writing and Accomplishments

- **Cryptanalysis Article:** J. de Mello Bezerra and Y. El Housni, "Smashing the Tower Number Field Sieve — BLS12-381 (Part 1)," juliodemellobezerra.github.io (2026).
- **Cryptanalysis Article:** Y. El Housni and J. de Mello Bezerra, "Beyond the Cheon-Kozaki-Kutsuma-Matsuo attack," yelhousni.github.io/beyond-kkm (2026).
- **Technical Article:** J. de Mello Bezerra, "Constructing Strong Pseudoprimes," juliodemellobezerra.github.io (2026).
- **Mathematics Preprint:** J. de Mello Bezerra, "Shintani cocycle decomposition of topological polylogarithms" arXiv:2610.32363 (2026).
- **Award:** Silver Medal (30th out of 9,531 participants), Brazilian Computer Science Olympics (2010).